

Nano-Thermite in the 9/11 Dust

What was actually found, who found it, and why the scientific mainstream rejected the explanation.

BOTTOM LINE

Real particles were found. The "explosive" reading of them was not. The whole claim rests on one 2009 paper published in a pay-to-publish journal, never replicated in a credible venue, and the one serious independent test identified the material as anti-corrosion primer paint. The claim survives on repetition, not on evidence.

01 The claim in one paragraph

In April 2009, a team led by Danish chemist Niels Harrit published *Active Thermitic Material Discovered in Dust from the 9/11 World Trade Center Catastrophe*.^[1] They collected World Trade Center dust from four separate Manhattan locations — one sample taken about ten minutes after the second tower fell — and found small red-and-gray chips in every one. Under electron microscopy the red layer showed iron-oxide grains around 100 nanometers across and aluminum in thin plate-like shapes. Their conclusion: this is unreacted **nano-thermite**, an engineered incendiary, and its presence means the towers were rigged.^[2] Architects & Engineers for 9/11 Truth has promoted the finding ever since as the physical smoking gun.^[3]

02 Quick verdict table

CLAIM	HOLDS UP?	WHY
Odd red/gray chips exist in WTC dust	YES	Independently confirmed. Nobody disputes the chips are there.
The paper was peer-reviewed	TECHNICALLY	Bentham's <i>Open Chemical Physics Journal</i> — pay-to-publish, no track record. The editor-in-chief resigned over it. ^[4]
The red layer is thermite	NO	Thermite needs free metallic aluminum. Independent analysis found none — the aluminum is locked up in clay. ^[5]
The findings were replicated	NO	Never reproduced in any peer-reviewed journal in 17 years.
Iron microspheres prove thermite	NO	Hot fires, welding, and cutting all make tiny iron spheres routinely. ^[6]
Thermite brought down the towers	NO	NIST found no evidence of explosives or incendiaries in any of the three buildings. ^[7]

03 Problem one: where it was published

The paper did not appear in a forensic, materials-science, or fire-engineering journal. It appeared in a Bentham open-access title that charges authors to publish and had essentially no reputation for review quality. The journal's editor-in-chief, Marie-Paule Pileni, resigned after it ran.^[4] Supporters call the resignation political pressure and note that a named reviewer later stood by his review.^[8] Critics call it an editor refusing to own a paper she says never cleared her desk. Either way, the paper skipped the venues where forensic chemists would have torn into it.

04 Problem two: the chemistry looks exactly like paint

This is the substantive objection, and it landed within days of publication.^[5]

- **The hexagonal plates** the authors read as engineered aluminum are consistent with **kaolin clay** — an aluminum silicate that is chemically inert, useless as thermite fuel, and a standard filler in paint.
- **The red pigment** is consistent with **hematite** — iron oxide, the thing that makes steel primer red in the first place.
- **The aluminum is bound, not free.** Thermite works because loose aluminum metal steals oxygen from iron oxide. Aluminum already bonded to silicon and oxygen in clay cannot do that. It is fuel that has already burned.
- **A specific paint match was found.** Analysts traced the recipe to the shop primer used by LaClede Steel, which fabricated the floor trusses — a formulation that does contain aluminum silicate and matches the published spectra

closely.^[5]

05 Problem three: the one real replication attempt

In 2012, forensic microscopist James Millette examined red/gray chips from archived WTC dust using SEM-EDS, FTIR, solvent soaking, and ashing.^[9] His finding, in plain terms: the chips are carbon steel coated with an epoxy resin pigmented with iron oxide and kaolin clay. Paint on steel. He reported no elemental aluminum particles of any size, and concluded the red layer is therefore not thermite or nano-thermite.

The counter-argument, stated fairly

Harrit's side has two real complaints, and they should not be waved away.^[10] First, Millette never ran the DSC ignition test — the calorimetry that was the entire basis of the "energetic" claim. Second, they argue he analyzed ordinary paint chips rather than the specific chips in question, since the dust contains several types. Co-author Kevin Ryan claimed to hold both kinds and to be able to tell them apart, but never released samples for testing.^[10] Millette also never published his work in a peer-reviewed journal as promised, which truther writers reasonably point to.^[11]

None of that rescues the claim. The compositional finding stands on its own: if there is no free aluminum, the material cannot be thermite regardless of what a calorimeter says. And the burden of proof runs toward the people asserting that engineered explosives were secretly installed in three skyscrapers — not toward the people asking them to show their samples.

06 Problem four: the story doesn't work physically

- **Thermite doesn't stop.** It carries its own oxygen and burns to completion — it can burn underwater. "Unreacted thermite" sitting loose in dust after a fire and collapse is chemically strange.^[6]
- **The melted steel isn't there.** Thermitic demolition of that scale would have left large pools of melted structural steel at Ground Zero. Nothing like that was found.^[6]
- **Thermite is a bad demolition tool.** NIST assessed it directly: thermite burns slowly and would need to sit against a massive steel section for minutes to weaken it — not how you drop a building.^[7]
- **The prep would be visible.** Rigging columns means stripping fireproofing and enclosures, torch-cutting, and re-covering the work — in occupied, secured buildings, undetected.^[12]
- **The seismic record is clean.** No signals before either tower began to fall.^[13]

07 Where it stands in 2026

Seventeen years on, the nano-thermite finding has not entered mainstream materials science or forensic engineering. It has never been replicated in a journal of standing. The closest independent analysis called it paint. NIST's investigations found no corroborating evidence for controlled demolition in the towers or WTC 7.^[14]

The claim circulates almost entirely inside the same advocacy ecosystem that produced it — AE911Truth, Global Research, 911Blogger, and a rotating cast of Substack revisionists. Recent 2025–2026 write-ups recycling the red-chip argument originate from that ecosystem, not from independent labs.^[15]

The interesting question is no longer chemical. It is why a single 2009 paper, published in a journal nobody reads and contradicted by the one person who tried to check it, has functioned for seventeen years as an unmovable evidentiary anchor. That is a story about how claims acquire durability, not about aluminum.

08 Sources

1. Harrit, N. et al. (2009). *Active Thermitic Material Discovered in Dust from the 9/11 World Trade Center Catastrophe*. The Open Chemical Physics Journal, Vol. 2, 7–31. benthamopenarchives.com/contents/pdf/TOCPJ/TOCPJ-2-7.pdf
2. Reprint and abstract, Global Research. globalresearch.ca/active-thermitic-material-discovered-in-dust-from-the-9-11-world-trade-center-catastrophe/13049
3. Architects & Engineers for 9/11 Truth, "High-Temperature Thermitic Reactions." ae911truth.org/evidence/high-temperature-thermitic-reactions
4. Dunbar & Reagan / Skeptical Inquirer (2011), "The 9/11 Truth Movement: The Top Conspiracy Theory, a Decade Later" — covers the Pileni resignation. skepticalinquirer.org/2011/07/the-9-11-truth-movement-the-top-conspiracy-theory-a-decade-later/
5. Metabunk technical thread on kaolin plates, hematite pigment, and the LaClede / Tnemec primer matching. metabunk.org/threads/investigating-active-thermitic-material-discovered-in-dust-from-the-9-11-wtc-catastrophe.9485/
6. Skeptical Inquirer, as above — on unreacted thermite, absent molten steel pools, and iron microspheres.
7. NIST, WTC Towers Investigation FAQ (Q29–30) on thermite, explosives testing, and seismic data. [nist.gov/world-trade-center-investigation/study-faqs/wtc-towers-investigation](https://www.nist.gov/world-trade-center-investigation/study-faqs/wtc-towers-investigation)
8. Harrit's response to the resignation and defenses of the paper: foreignpolicyjournal.com/2013/12/09/how-to-debunk-wtc-thermite/ and opednews.com — "Why the Harrit Nano-thermite paper has not yet been debunked"
9. Millette, J. (2012). *Progress Report on the Analysis of Red/Gray Chips in WTC Dust*, MVA Scientific Consultants. aneta.org/911experiments_com/Millette/paper/index.htm

- .0. Harrit-side rebuttal to Millette, including the Kevin Ryan sample dispute and the missing DSC test. 911debunkers.blogspot.com/2014/06/how-to-distinguish-thermitic-redgray.html
- .1. Off-Guardian (2016), "Jones, Harrit, Mohr, Millette — the red-gray chips" — on Millette's failure to publish. off-guardian.org/2016/09/25/jones-harrit-mohr-millette-the-red-gray-chips/
- .2. NIST, WTC 7 Investigation FAQ (Q34–36) on thermite feasibility and demolition prep. nist.gov/world-trade-center-investigation/study-faqs/wtc-7-investigation
- .3. NIST WTC Towers FAQ, seismic record — see [7].
- .4. NIST, About the WTC Disaster Investigation — overall findings on alternative hypotheses. nist.gov/world-trade-center-investigation/about-investigation
- .5. Example of continued 2026 circulation within the advocacy ecosystem. 911revision.substack.com/p/reaction-nanothermite-myth-busted

Prepared 18 July 2026. Sources include both proponent and critic material by design; proponent links are included so the primary claims can be read in their own words rather than summarized secondhand.